Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/23/2005
PI Name:	Krauzlis, Rich Ph.D.		
Project Title:	Investigation of Neural Strategies of Visual Search		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rich@salk.edu	**Fax:**	FY 858-546-8526
PI Organization Type:	NON-PROFIT	**Phone:**	858-453-4100 1257
Organization Name:	Salk Institute		
PI Address 1:	10010 North Torrey Pines Road		
PI Address 2:			
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92037	**Congressional District:**	50
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	12/09/2002	**End Date:**	12/09/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1465		
Performance Goal No.:			
Performance Goal Text:			
	Human performance is easily impaired in a variety of aerospace tasks – such as remotely docking a spacecraft, or flying an aircraft in poor visibility, or finding and tracking a particular target airplane on a crowded air-traffic control display. These impairments occur because aerospace environments can degrade or remove the sensory cues used to drive and coordinate voluntary actions, and display systems may not be optimally matched to the human user. Fortunately, much progress has been made recently in the field of neuroscience toward understanding the physiological mechanisms responsible for voluntary motor behaviors. In particular, we are beginning to understand how the brain accomplishes visual search – the process that allows us to quickly respond to important items in our surroundings, even in the presence of other irrelevant stimuli. Understanding how individual neurons participate in this process is critical for identifying the factors that limit the speed and accuracy of visual search. Knowledge of these neural factors could be used to design visual displays that conform to the temporal and spatial constraints imposed by human neurobiology, thereby directly improving human performance in man-machine interfaces and virtual environments.		

Arguably the most important class of movements made by primates during visual search are eye movements, because control of the position of the eyes determines which visual information is sampled and for how long. Decades of anatomical and physiological research have provided a detailed outline of the brain pathways that are responsible for the control of eye movements in primates. These pathways involve a network of areas including visual areas of the cerebral cortex (which provide information about the content of the visual scene), motor regions of the cerebellum and brainstem (which form the final motor commands), and several regions that lie in-between (and which possess a mixture of sensory and motor properties). It is these latter, intermediate, brain regions that are most likely responsible for forming the decision about where the eyes should move next during visual search, because they lie at the interface between the visual areas that provide evidence about the location of stimuli and the motor regions that execute the movements. One of these structures, the superior colliculus (SC), has long been known to play a pivotal role in the control of saccadic eye movements. However, most studies of the SC have focused on its contribution to the motor execution of saccades. Because of the importance of the SC in the decision processes that precede saccade generation, examination of this structure during visual search is likely to reveal fundamental aspects of the visual neural computations involved in identifying, localizing, and selecting visual targets. The main goal of this project would be to measure how neurons in the superior colliculus (SC) change their activity during a visual search task. Specifically, we propose to measure how the activity of these neurons is altered by the discriminability of visual targets and test whether these changes can predict the changes in the subjects' performance. Understanding how the information encoded by these neurons constrains overall search performance will foster the development of more predictive models of human performance and ultimately lead to the design of better aerospace displays and interfaces. Goal: To develop a computational model of decision-making, involving multiple sources, agents and outputs, based on neurophysiological data.

Task Description:

The phenomenon: Visual search is a basic form of decision-making that is performed by humans and non-human primates several times a second (e.g., as you read this text). The brain pathways that accomplish this behavior have been identified, but the processing steps and computations that are involved in supporting this phenomenon are not known.

The experimental approach: By recording from single neurons in the brains of non-human primates as they perform visual search, we have a window into the internal processing steps that support visual search and how these change over time on a millisecond time scale – data that cannot be obtained from behavioral experiments with humans. In addition, we conduct parallel behavioral experiments with humans to confirm the applicability of this approach.

The hypothesis: By recording the activity of neurons during visual search, we can identify the biological computations that underlie performance in visual-motor tasks and decision-making.

Countermeasure readiness level: 2 at start of project, 3 now. The original hypothesis has now been validated and we are in the process of understanding in detail the neural processes steps that support neurally-mediated visual search. Specifically, we are testing how different sources of information are combined and read out by neurons during this basic decision task.

Expected countermeasure readiness level at end of project: 4 to 5. Once we understand the dynamic regulation of top-down and bottom-up factors, we will have identified the key features that are relevant for constructing biologically inspired models of visual search. We will then formulate how these features should be incorporated into computer-based models of decision-making (level 4), and then prove the feasibility and efficacy of this approach by constructing a model of visual search based on these biological features (level 5). After this initial demonstration, the concept may be applied to other processes that would benefit from automated decision-making (levels 5 – 6). The results from these experiments provide new insights into the biological mechanisms underlying visual search behavior in humans and other primates. Although the immediate goals of the project were to expand our basic biological knowledge, the results are also be applicable to the stated goals of NASA's Space Human Factors, Intelligent Systems, and Aerospace Operation Systems Programs. Biological systems are capable of accomplishing tasks that have proved extremely difficult to duplicate using artificial intelligence and robotic systems – especially the job of detecting, discriminating and responding to stimuli quickly and accurately, even in the presence of background noise. Our results have provided new data about the algorithms and information processing streams used by biological systems to accomplish such tasks, and may therefore aid in the development of robust computational models of human visuomotor performance and in the design of the autonomous systems for future aerospace applications.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

1) Analysis of neuronal activity during visual search when prior information is provided : increased sensitivity or weighted evidence? We have continued and extended our analysis of neuronal activity in the SC of two subjects while performing a visual search and discrimination task. The data that are the basis of this analysis were obtained during the previous review period. In brief, we recorded from saccade-related neurons (n=85) in the SC of two subjects during a luminance discrimination task. Subjects initially fixated a central fixation cross on a random noise background, and two boxes at 6 deg on either side of fixation demarcated possible target locations. On cued trials one box briefly reversed contrast, indicating the likely location of the upcoming target (75% valid). After a random interval, two Gaussian-blurred discs were added, with a luminance increment (SNR = 0 to 4) added to one disk. The subject was rewarded for making a saccade to the brighter disc. This new analysis is aimed at addressing a basic question about how prior information affects performance during visual search: does it change the sensitivity to the visual input itself or does it weight the visual inputs without changing the underlying evidence? The former explanation is the traditional interpretation, whereas the latter explanation is a more recent interpretation based on signal detection theory and applications of Bayesian inference. The results of our analysis favor the weighted evidence interpretation. The subjects' performance in the task showed a clear cue validity effect higher percent correct when the cue correctly indicated the target location. Neurons in the superior colliculus showed a neural correlate of this cueing effect higher activity on validly cued trials compared to invalidly cued trials. However, ROC analysis indicated that this effect was almost entirely due to a bias in activity caused by the presence or absence of the cue in the neuron s response field. The ability of the neuron to visually discriminate between target and distractor was largely unaltered by the presence or absence of a cue. These findings are consistent with the idea that saccadic choice behavior is guided by a weighted integration of sensory evidence, and that the activity we recorded in the superior colliculus lies after the point at which those weights are applied.
2) Testing the contributions made by the SC and the FEF during a visual search task when the instructions change Most of our effort during this review period was devoted to training subjects on a new visual search task in which the

Task Progress:	instructions change unpredictably. The significance of this approach is that visual search is often performed under changing task demands, but the neural mechanisms responsible for performance monitoring are largely unknown. By testing the two major brain regions involved in visual search under these conditions, we expect to identify key properties of the underlying neuronal mechanisms. Our working hypothesis is that changes in visual search caused by “bottom-up” changes in visual signals will be largely mediated by the superior colliculus, whereas changes in visual search caused by “top-down” changes in cognitive instructions will be largely mediated by the frontal eye fields in the cerebral cortex. To test this idea, we have trained a subject to perform a task that requires him to change the goal of their movement based on either a visual or symbolic instruction. Using an analysis procedure based on signal detection theory, we will identify when neurons discriminate the two types of instructions. Our prediction is that SC neurons will discriminate visual instructions sooner than FEF neurons, but that FEF neurons will discriminate symbolic instructions sooner than SC neurons. We are now in the final stages of training the first subject and expect to begin recording within the next 60 days. By the next reporting period, we expect to have sufficient neuronal data to provide a clear test of our hypothesis. 3) Extending our behavioral studies of visual search Based on the feedback and comments received during the Bioastronautics Investigator’s Workshop in Galveston Texas in January 2005, we have now initiated an additional set of studies aimed at making our neurophysiological studies more directly relevant to issues of human performance. These studies address a major gap in our understanding of search – very little is known about the planning of eye movements that are guided by non-visual representations of where to look next. Although non-visual representations may at first seem an unlikely basis for visual search, there are many conditions under which search movements are planned and guided with uncertain or minimal visual information. The goal of these studies is to examine the behavioral properties of this class of search in human subjects, and then identify the underlying neural mechanisms. In the initial set of experiments, human subjects searched for a target on either a mean grey screen or a structured noise background (1/f, pink noise). The location of the target in each case did not correspond to any visual element on the screen, but was drawn from a probability distribution with a given center and spread. The subjects were asked to find the target as quickly as possible; an eye movement to the correct location was rewarded with a tone. We have preliminary data from four subjects, including two naive to the purposes of the experiment. Following practice, subjects' eye movements revealed that they had learned both the location of the center of the probability distribution from which the targets were drawn, and later in the session information about its spread. Different eye movement scanning strategies adopted by individual subjects led to differential learning for the pink noise background versus the grey background. Subjects seemed to utilize the visual landmarks in the pink noise background despite the fact that they were uncorrelated with target location. These early results show that humans are able to build estimates of where to look even when visual cues did not provide extra information. This task offers a method for probing how prior information is integrated with actual visual information within a Bayesian framework, an approach we plan to extend and exploit during the upcoming year. Summary of results to date: The first half of the project has been completed. We have recorded the activity of single neurons in two non-human primates as they perform a visual search task. We manipulated two factors related to the non-human primates’ decision: 1) prior information about the likely location of the target, and 2) the strength of the sensory signals specifying the target. The key scientific findings from this phase of the project are: · Individual neurons are sensitive to both prior information and sensory signals but at low “efficiency” compared to the subjects’ behavior, indicating that the choice is guided by pooling information over many neural elements. · The sensitivity of neurons to the two factors (prior information, sensory information) is not strictly additive, but instead changes within a trial. · The interaction between the prior information and sensory information on single neurons is related to the behavioral strategy of the subject, and possibly their recent history of choices. · The changes in performance caused by prior information are not wholly accounted for by the activity of these neurons, indicating that the sensory information is complemented by activity from another brain area (probably the frontal eye fields).
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Abstracts for Journals and Proceedings	Krauzlis, R.J.; Liston, D.; Eckstein, M.P. "Cueing effects on superior colliculus neurons: increased sensitivity or weighted evidence?" Neurosciences Abstracts , 2005. Neurosciences Abstracts 31, in press 2005. , Nov-2005
Abstracts for Journals and Proceedings	Chukoskie, L.; Schwartz, O.; Senjowski, T.; Dayan, P.; Krauzlis, R.J. "Deciding where to look when there’s not much to see." Neurosciences Abstracts, 2005. Neurosciences Abstracts 31, in press, 2005. , Nov-2005
Abstracts for Journals and Proceedings	Nummela, S.U.; Lovejoy, L.P.; Krauzlis, R.J. "Saccade selection when target probability is dynamically manipulated." Neurosciences Abstracts 2005. Neurosciences Abstracts .31, in press 2005. , Nov-2005
Articles in Peer-reviewed Journals	Liston D, Krauzlis RJ. "Shared decision signal explains performance and timing of pursuit and saccadic eye movements." Journal of Vision, in press, 2005. , Jun-2005